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2.2.2 Component in the direction of ...

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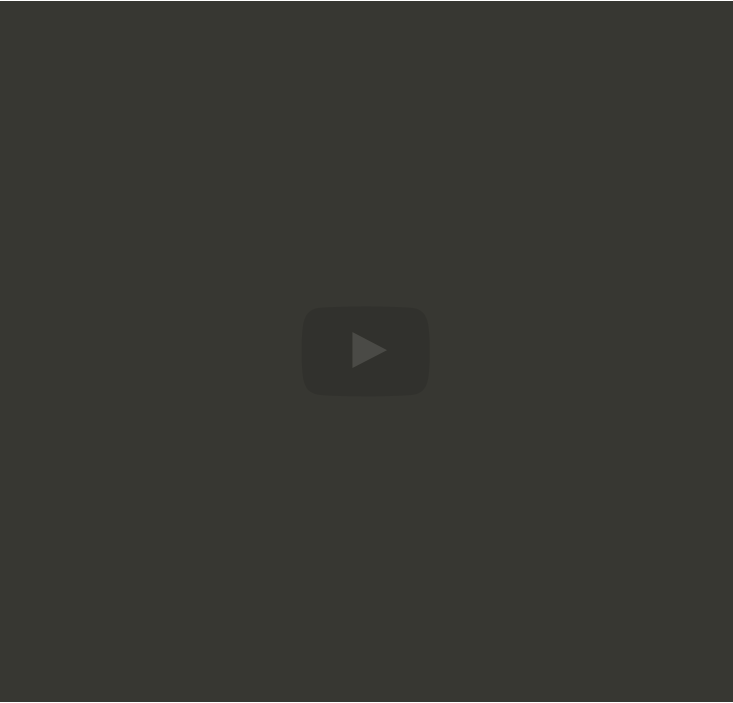
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Which we can then rearrange as - let's see - the scalar times a vector is the same as a vector times a scalar. So we get a times a transpose b divided by a transpose a. And then we can place some clever parentheses. And we get a times a transpose divided by a transpose a times b. And why do I like that so much? I like it so much because this tells me what the matrix is such that when I apply it to any vector I get the component of

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Reading assignment

1/1 point (graded)



Read Unit 2.2.2 of the notes. [\[LINK\]](#)

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Homework 2.2.2.1.

1/1 point (graded)

Let $\boldsymbol{a} \in \mathbb{C}^m$.

$$\left(\frac{\boldsymbol{a}\boldsymbol{a}^H}{\boldsymbol{a}^H\boldsymbol{a}}\right)\left(\frac{\boldsymbol{a}\boldsymbol{a}^H}{\boldsymbol{a}^H\boldsymbol{a}}\right)=\frac{\boldsymbol{a}\boldsymbol{a}^H}{\boldsymbol{a}^H\boldsymbol{a}}.$$

ALWAYS

 Answer: ALWAYS

Interpret what this means about a matrix that projects onto a vector.

Solution:

Now prove it!

For complete solution, see [[LINK](#)].

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Homework 2.2.2.2

1/1 point (graded)
Let $\boldsymbol{a} \in \mathbb{C}^m$.

$$\left(\frac{\boldsymbol{a}\boldsymbol{a}^H}{\boldsymbol{a}^H\boldsymbol{a}}\right)\left(\boldsymbol{I}-\frac{\boldsymbol{a}\boldsymbol{a}^H}{\boldsymbol{a}^H\boldsymbol{a}}\right)=\boldsymbol{0}.$$

ALWAYS

 Answer: ALWAYS

Interpret what this means.

Solution:

Now prove it!

For complete solution, see [[LINK](#)].

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Homework 2.2.2.3

1/1 point (graded)

Let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}^n$, $\hat{\boldsymbol{b}} = \frac{\boldsymbol{a}\boldsymbol{a}^H}{\boldsymbol{a}^H\boldsymbol{a}}\boldsymbol{b}$, and $\boldsymbol{b}^\perp = \boldsymbol{b} - \hat{\boldsymbol{b}}$.

$$\hat{\boldsymbol{b}}^H\boldsymbol{b}^\perp=0.$$

ALWAYS

 Answer: ALWAYS

Interpret what this means.

Solution:

Now prove it!

For complete solution, see [[LINK](#)].

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